

SECTION 33 05 00

COMMON WORK RESULTS FOR UTILITIES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Piping joining materials.
 - 2. Dielectric fittings.
 - 3. Sleeves.
 - 4. Identification devices.
 - 5. Grout.
 - 6. Piping system common requirements.
 - 7. Equipment installation common requirements.
 - 8. Concrete bases.
 - 9. Metal supports and anchorages.

1.2 DEFINITIONS

- A. Exposed Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions.
- B. Concealed Installations: Concealed from view and protected from weather conditions and physical contact by building occupants but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.

1.3 SUBMITTALS

- A. Product Data: For the following:
 - 1. Dielectric fittings.
 - 2. Identification devices.
- B. Welding certificates.

1.4 QUALITY ASSURANCE

- A. Steel Support Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- B. Steel Piping Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."

1. Comply with provisions in ASME B31 Series, "Code for Pressure Piping."
 2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.
- C. Comply with ASME A13.1 for lettering size, length of color field, colors, and viewing angles of identification devices.

PART 2 - PRODUCTS

2.1 PIPING JOINING MATERIALS

- A. Pipe-Flange Gasket Materials: Suitable for chemical and thermal conditions of piping system contents.
1. ASME B16.21, nonmetallic, flat, asbestos free, 1/8-inch maximum thickness, unless otherwise indicated.
 - a. Full-Face Type: For flat-face, Class 125, cast-iron and cast-bronze flanges.
 - b. Narrow-Face Type: For raised-face, Class 250, cast-iron and steel flanges.
 2. AWWA C110, rubber, flat face, 1/8 inch thick, unless otherwise indicated; and full-face or ring type, unless otherwise indicated.
- B. Flange Bolts and Nuts: ASME B18.2.1, carbon steel, unless otherwise indicated.
- C. Plastic, Pipe-Flange Gasket, Bolts, and Nuts: Type and material recommended by piping system manufacturer, unless otherwise indicated.
- D. Solder Filler Metals: ASTM B 32, lead-free alloys. Include water-flushable flux according to ASTM B 813.
- E. Brazing Filler Metals: AWS A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing, unless otherwise indicated; and AWS A5.8, BAg1, silver alloy for refrigerant piping, unless otherwise indicated.
- F. Welding Filler Metals: Comply with AWS D10.12/D10.12M for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.
- G. Solvent Cements for Joining Plastic Piping:
1. ABS Piping: ASTM D 2235.
 2. CPVC Piping: ASTM F 493.
 3. PVC Piping: ASTM D 2564. Include primer according to ASTM F 656.
 4. PVC to ABS Piping Transition: ASTM D 3138.
- H. Fiberglass Pipe Adhesive: As furnished or recommended by pipe manufacturer.

2.2 DIELECTRIC FITTINGS

- A. Dielectric Fittings, General: Assembly of copper alloy and ferrous materials or ferrous material body with separating nonconductive insulating material suitable for system fluid, pressure, and temperature.
- B. Dielectric Unions:
 - 1. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Capitol Manufacturing Co.
 - b. Central Plastics Company.
 - c. Epco Sales, Inc.
 - d. Hart Industries, International, Inc.
 - e. Watts Water Technologies, Inc.
 - f. Zurn Plumbing Products Group; Wilkins Div.
 - 3. Description: Factory fabricated, union, NPS 2 and smaller.
 - a. Pressure Rating: 150 psig to 250 psig at 180 deg F.
 - b. End Connections: Solder-joint copper alloy and threaded ferrous; threaded ferrous.
- C. Dielectric Flanges:
 - 1. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Capitol Manufacturing Co.
 - b. Central Plastics Company.
 - c. Epco Sales, Inc.
 - d. Watts Water Technologies, Inc.
 - 3. Description: Factory-fabricated, bolted, companion-flange assembly, NPS 2-1/2 to NPS 4 and larger.
 - a. Pressure Rating: 150 psig to 300 psig.
 - b. End Connections: Solder-joint copper alloy and threaded ferrous; threaded solder-joint copper alloy and threaded ferrous.
- D. Dielectric Couplings:

1. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Calpico, Inc.
 - b. Lochinvar Corporation.
3. Description: Galvanized-steel coupling with inert and noncorrosive, thermoplastic lining, NPS 3 and smaller.
 - a. Pressure Rating: 300 psig at 225 deg F.
 - b. End Connections: Threaded.

E. Dielectric Nipples:

1. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Perfection Corporation.
 - b. Precision Plumbing Products, Inc.
 - c. Victaulic Company.
3. Description: Electroplated steel nipple with inert and noncorrosive, thermoplastic lining.
 - a. Pressure Rating: 300 psig at 225 deg F.
 - b. End Connections: Threaded or grooved.

2.3 SLEEVES

- A. Mechanical sleeve seals for pipe penetrations are specified in Division 22 Section "Common Work Results for Plumbing."
- B. Galvanized-Steel Sheet Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.
- C. Steel Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, galvanized, plain ends.
- D. Cast-Iron Sleeves: Cast or fabricated "wall pipe" equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop, unless otherwise indicated.
- E. Molded PVC Sleeves: Permanent, with nailing flange for attaching to wooden forms.
- F. PVC Pipe Sleeves: ASTM D 1785, Schedule 40.
- G. Molded PE Sleeves: Reusable, PE, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.

2.4 IDENTIFICATION DEVICES

- A. Equipment Nameplates: Metal permanently fastened to equipment with data engraved or stamped.
 - 1. Data: Manufacturer, product name, model number, serial number, capacity, operating and power characteristics, labels of tested compliances, and essential data.
 - 2. Location: Accessible and visible.
- B. Snap-on Plastic Pipe Markers: Manufacturer's standard preprinted, semirigid, snap-on type. Include color-coding according to ASME A13.1, unless otherwise indicated.
- C. Pressure-Sensitive Pipe Markers: Manufacturer's standard preprinted, color-coded, pressure-sensitive-vinyl type with permanent adhesive.
- D. Pipes with OD, Including Insulation, Less Than 6 Inches: Full-band pipe markers, extending 360 degrees around pipe at each location.
- E. Pipes with OD, Including Insulation, 6 Inches and Larger: Either full-band or strip-type pipe markers, at least three times letter height and of length required for label.
- F. Lettering: Use piping system terms indicated and abbreviate only as necessary for each application length.
 - 1. Arrows: Either integrally with piping system service lettering to accommodate both directions of flow, or as separate unit on each pipe marker to indicate direction of flow.
- G. Plastic Tape: Manufacturer's standard color-coded, pressure-sensitive, self-adhesive vinyl tape, at least 3 mils thick.
 - 1. Width: 1-1/2 inches on pipes with OD, including insulation, less than 6 inches; 2-1/2 inches for larger pipes.
 - 2. Color: Comply with ASME A13.1, unless otherwise indicated.
- H. Valve Tags: Stamped or engraved with 1/4-inch letters for piping system abbreviation and 1/2-inch sequenced numbers. Include 5/32-inch hole for fastener.
 - 1. Material: 0.032-inch- thick, polished brass or aluminum.
 - 2. Material: 0.0375-inch- thick stainless steel.
 - 3. Material: 3/32-inch- thick plastic laminate with 2 black surfaces and a white inner layer.
 - 4. Material: Valve manufacturer's standard solid plastic.
 - 5. Size: 1-1/2 inch in diameter, unless otherwise indicated.
 - 6. Shape: As indicated for each piping system.
- I. Valve Tag Fasteners: Brass, wire-link or beaded chain; or brass S-hooks.
- J. Engraved Plastic-Laminate Signs: ASTM D 709, Type I, cellulose, paper-base, phenolic-resin-laminate engraving stock; Grade ES-2, black surface, black phenolic core, with white melamine subcore, unless otherwise indicated. Fabricate in sizes required for message. Provide holes for mechanical fastening.

1. Engraving: Engraver's standard letter style, of sizes and with terms to match equipment identification.
 2. Thickness: 1/8 inch, unless otherwise indicated.
 3. Thickness: 1/16 inch, for units up to 20 sq. in. or 8 inches in length, and 1/8 inch for larger units.
 4. Fasteners: Self-tapping, stainless-steel screws or contact-type permanent adhesive.
- K. Plastic Equipment Markers: Manufacturer's standard laminated plastic, in the following color codes:
1. Green: Cooling equipment and components.
 2. Yellow: Heating equipment and components.
 3. Brown: Energy reclamation equipment and components.
 4. Blue: Equipment and components that do not meet criteria above.
 5. Hazardous Equipment: Use colors and designs recommended by ASME A13.1.
 6. Terminology: Match schedules as closely as possible. Include the following:
 - a. Name and plan number.
 - b. Equipment service.
 - c. Design capacity.
 - d. Other design parameters such as pressure drop, entering and leaving conditions, and speed.
 7. Size: 2-1/2 by 4 inches for control devices, dampers, and valves; 4-1/2 by 6 inches for equipment.

2.5 GROUT

- A. Description: ASTM C 1107, Grade B, nonshrink and nonmetallic, dry hydraulic-cement grout.
1. Characteristics: Post hardening, volume adjusting, nonstaining, noncorrosive, nongaseous, and recommended for interior and exterior applications.
 2. Design Mix: 5000-psi, 28-day compressive strength.
 3. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 DIELECTRIC FITTING APPLICATIONS

- A. Dry Piping Systems: Connect piping of dissimilar metals with the following:
1. NPS 2 and Smaller: Dielectric unions.
 2. NPS 2-1/2 and Larger: Dielectric flanges.
- B. Wet Piping Systems: Connect piping of dissimilar metals with the following:
1. NPS 2 and Smaller: Dielectric couplings or dielectric nipples.
 2. NPS 2-1/2 and Larger: Dielectric nipples.

3.2 PIPING INSTALLATION

- A. Install piping according to the following requirements and Division 33 Sections specifying piping systems.
- B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on the Coordination Drawings.
- C. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- D. Install piping to permit valve servicing.
- E. Install piping at indicated slopes.
- F. Install piping free of sags and bends.
- G. Install fittings for changes in direction and branch connections.
- H. Select system components with pressure rating equal to or greater than system operating pressure.
- I. Sleeves are not required for core-drilled holes.
- J. Permanent sleeves are not required for holes formed by removable PE sleeves.
- K. Install sleeves for pipes passing through concrete and masonry walls and concrete floor and roof slabs.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of equipment areas or other wet areas 2 inches above finished floor level.
 - 2. Install sleeves in new walls and slabs as new walls and slabs are constructed.
 - a. PVC or Steel Pipe Sleeves: For pipes smaller than NPS 6.
 - b. Steel Sheet Sleeves: For pipes NPS 6 and larger, penetrating gypsum-board partitions.
- L. Verify final equipment locations for roughing-in.
- M. Refer to equipment specifications in other Sections for roughing-in requirements.

3.3 PIPING JOINT CONSTRUCTION

- A. Join pipe and fittings according to the following requirements and Division 33 Sections specifying piping systems.

- B. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
- C. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.
- D. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
 - 2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.
- E. Welded Joints: Construct joints according to AWS D10.12/D10.12M, using qualified processes and welding operators according to Part 1 "Quality Assurance" Article.
- F. Flanged Joints: Select appropriate gasket material, size, type, and thickness for service application. Install gasket concentrically positioned. Use suitable lubricants on bolt threads.
- G. Grooved Joints: Assemble joints with grooved-end pipe coupling with coupling housing, gasket, lubricant, and bolts according to coupling and fitting manufacturer's written instructions.
- H. Soldered Joints: Apply ASTM B 813 water-flushable flux, unless otherwise indicated, to tube end. Construct joints according to ASTM B 828 or CDA's "Copper Tube Handbook," using lead-free solder alloy (0.20 percent maximum lead content) complying with ASTM B 32.
- I. Brazed Joints: Construct joints according to AWS's "Brazing Handbook," "Pipe and Tube" Chapter, using copper-phosphorus brazing filler metal complying with AWS A5.8.
- J. Pressure-Sealed Joints: Assemble joints for plain-end copper tube and mechanical pressure seal fitting with proprietary crimping tool to according to fitting manufacturer's written instructions.
- K. Plastic Piping Solvent-Cemented Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:
 - 1. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.
 - 2. ABS Piping: Join according to ASTM D 2235 and ASTM D 2661 appendixes.
 - 3. CPVC Piping: Join according to ASTM D 2846/D 2846M Appendix.
 - 4. PVC Pressure Piping: Join schedule number ASTM D 1785, PVC pipe and PVC socket fittings according to ASTM D 2672. Join other-than-schedule-number PVC pipe and socket fittings according to ASTM D 2855.
 - 5. PVC Nonpressure Piping: Join according to ASTM D 2855.
 - 6. PVC to ABS Nonpressure Transition Fittings: Join according to ASTM D 3138 Appendix.
- L. Plastic Pressure Piping Gasketed Joints: Join according to ASTM D 3139.
- M. Plastic Nonpressure Piping Gasketed Joints: Join according to ASTM D 3212.

- N. Plastic Piping Heat-Fusion Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join according to ASTM D 2657.
 - 1. Plain-End PE Pipe and Fittings: Use butt fusion.
 - 2. Plain-End PE Pipe and Socket Fittings: Use socket fusion.
- O. Bonded Joints: Prepare pipe ends and fittings, apply adhesive, and join according to pipe manufacturer's written instructions.

3.4 PIPING CONNECTIONS

- A. Make connections according to the following, unless otherwise indicated:
 - 1. Install unions, in piping NPS 2 and smaller, adjacent to each valve and at final connection to each piece of equipment.
 - 2. Install flanges, in piping NPS 2-1/2 and larger, adjacent to flanged valves and at final connection to each piece of equipment.
 - 3. Install dielectric fittings at connections of dissimilar metal pipes.

3.5 EQUIPMENT INSTALLATION

- A. Install equipment level and plumb, unless otherwise indicated.
- B. Install equipment to facilitate service, maintenance, and repair or replacement of components. Connect equipment for ease of disconnecting, with minimum interference with other installations. Extend grease fittings to an accessible location.
- C. Install equipment to allow right of way to piping systems installed at required slope.

3.6 IDENTIFICATION

- A. Piping Systems: Install pipe markers on each system. Include arrows showing normal direction of flow.
 - 1. Plastic markers, with application systems. Install on insulation segment if required for hot noninsulated piping.
 - 2. Locate pipe markers on exposed piping according to the following:
 - a. Near each valve and control device.
 - b. Near each branch, excluding short takeoffs for equipment and terminal units. Mark each pipe at branch if flow pattern is not obvious.
 - c. Near locations where pipes pass through walls or floors or enter inaccessible enclosures.
 - d. At manholes and similar access points that permit view of concealed piping.
 - e. Near major equipment items and other points of origination and termination.
- B. Equipment: Install engraved plastic-laminate sign or equipment marker on or near each major item of equipment.

1. Lettering Size: Minimum 1/4 inch high for name of unit if viewing distance is less than 24 inches, 1/2 inch high for distances up to 72 inches, and proportionately larger lettering for greater distances. Provide secondary lettering two-thirds to three-fourths of size of principal lettering.
 2. Text of Signs: Provide name of identified unit. Include text to distinguish among multiple units, inform user of operational requirements, indicate safety and emergency precautions, and warn of hazards and improper operations.
- C. Adjusting: Relocate identifying devices that become visually blocked by work of this or other Divisions.

3.7 CONCRETE BASES

- A. Concrete Bases: Anchor equipment to concrete base according to equipment manufacturer's written instructions and according to seismic codes at Project.
1. Construct concrete bases of dimensions indicated, but not less than 4 inches larger in both directions than supported unit.
 2. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch centers around the full perimeter of base.
 3. Install epoxy-coated anchor bolts for supported equipment that extend through concrete base, and anchor into structural concrete floor.
 4. Place and secure anchorage devices. Use supported equipment manufacturer's setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 5. Install anchor bolts to elevations required for proper attachment to supported equipment.
 6. Install anchor bolts according to anchor-bolt manufacturer's written instructions.
 7. Use 3000-psi, 28-day compressive-strength concrete and reinforcement as specified in Division 03 Section "Cast-in-Place Concrete/Miscellaneous Cast-in-Place Concrete."

3.8 ERECTION OF METAL SUPPORTS AND ANCHORAGES

- A. Refer to Division 05 Section "Metal Fabrications" for structural steel.
- B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor piped utility materials and equipment.
- C. Field Welding: Comply with AWS D1.1/D1.1M.

3.9 GROUTING

- A. Mix and install grout for equipment base bearing surfaces, pump and other equipment base plates, and anchors.
- B. Clean surfaces that will come into contact with grout.
- C. Provide forms as required for placement of grout.
- D. Avoid air entrapment during placement of grout.

- E. Place grout, completely filling equipment bases.
- F. Place grout on concrete bases and provide smooth bearing surface for equipment.
- G. Place grout around anchors.
- H. Cure placed grout.

END OF SECTION

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SECTION 33 10 00

WATER DISTRIBUTION SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

- A. Conditions of the contract (General, Supplementary, and other conditions), the General Requirements (Sections of Division One) and the Drawings Apply to the Work of this Section.
- B. Water Distribution System for Domestic Consumption and Fire Fighting as Indicated on The Drawings, Herein Specified, and Required for a Complete and Proper Installation.

1.2 WORK SPECIFIED HEREIN

- A. Provide all labor, materials, equipment, and services necessary to complete the work consisting of furnishing and install the complete distribution facilities, herein referred to as water mains and/or to modify existing water facilities as shown on Construction Documents and in accordance with the specifications and the Uniform Design and Construction Standards for Potable Water Distribution Systems, 3rd Edition (UDACS), all Clark County (CC) amendments and latest revisions. Where conflicts arise, UDACS and CC amendments and latest revisions shall govern.
- B. Unless otherwise indicated on the plans and/or specified in the Special Provisions, the construction of water mains shall include: excavation and backfill, construction of concrete structures, anchors, thrust blocks, supports, sleeving, and encasement; the furnishing, installing, testing, and disinfecting of water pipelines, fittings, valves, services, fire hydrants, and all appurtenances; the removal and/or restoration of existing facilities, and all work in accordance with the plans and specifications.

1.3 SUBMITTALS

- A. Submit under provisions of the specifications; subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site improvements, Clark County and respective Local Utility Companies.
- B. Product Data: Submit manufacturer's data for product characteristics, valves, fittings, insulation jointing materials, and all other materials required for a complete installation.
- C. Submit manufacturer's installation instructions and recommendations. Contractor shall submit all specified product literature and shop drawings in accordance with the requirements of these specifications, UDACS, and from the Las Vegas Valley Water District approved products list.
- D. Provide County's representative with data on pipe materials, pipe fittings, hydrants, valves, and accessories.

- E. Submit copy of results of meter tests and hydrostatic pressure test to CCSD's representative, utility company and project Civil Engineer upon completion of water distribution backfilling operations.
- F. Manufacturer's Certificate: Certify that products meet or exceed state and local requirements.
- G. Provide County's representative with a Disinfection report and Bacteriological report including dates, times, results, locations, personnel involved, certifications, etc.
- H. As-Builts: Accurately record actual installed locations of piping mains, valves, connections, and top of pipe elevations.
- I. Identify and describe unexpected variations to subsoil conditions and location of uncharted utilities.

1.4 QUALITY ASSURANCE

- A. Comply with the Uniform Standard Specifications for Public Work's Construction Off-Site Improvements, Nye County.
- B. Comply with the Uniform Design and Construction Standards for Potable Water Systems, 3rd edition.
- C. Comply with the Uniform Standard Drawings for Public Works Construction Off-site Improvements, Nye County.
- D. Construction of water main and building service pipelines utilizing plant fabricated pipe and other appurtenant materials, installed for conveyance of potable water. Relocation or adjustment of existing facilities as may be indicated on the drawings.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, handle, and protect products to site under provisions of Section 01 60 00.
- B. Deliver new packaged materials, well-marked and identified, clean, dry and protected against dampness, freezing and damage.
- C. Storage of PVC Materials:
 - 1. Store in unit packages as received from manufacturer until just prior to use
 - 2. Stack units in such manner as to prevent deformation to pipe barrel and bells
 - 3. Protect from direct sunlight with opaque material if storage period will exceed six (6) weeks.
- D. Avoid severe impact blows, gouging, or cutting by metal surfaces or rocks.

1.6 FIELD MEASUREMENTS

- A. Verify actual locations of water distribution system and other construction to which water distribution must fit by accurate field measurements before installation. Coordinate installation schedule with construction progress to avoid delay of work.

1.7 WARRANTY

- A. Provide one (1) year written warranty under provisions of specifications.
- B. Warranty: All materials and workmanship provided are guaranteed against defects after completion and final acceptance of the work. Defects due to faulty materials or workmanship developed during the guarantee period shall be satisfactorily repaired or replaced by the Contactor at his expense.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies; and subject to compliance with the requirements of the Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.

PART 3 - EXECUTION

3.1 INSPECTION AND RESPONSIBILITY OF MATERIALS

- A. During the process of unloading, all pipe and accessories shall be inspected by the Contractor for damage. Notify the Engineer of all material found that has cracks, flaws, or defects. The Engineer shall inspect the damaged material and have the right to reject any materials he finds unsatisfactory. Promptly remove all rejected material from the site.
- B. Pipe and other accessories shall, unless otherwise directed by the Engineer, be unloaded at the point of delivery, hauled to and distributed at the site of the project by the Contractor. They shall at all times be handled with care to avoid damage.
- C. The Contractor shall be responsible for all material furnished by him and shall replace at his own expense all such material that is found to be defective in manufacturer or that has become damaged in handling after delivery by the manufacturer. This shall include the furnishing of all material and labor required for the replacement of installed material discovered defective prior to the final acceptance of the work.
- D. The Contractor shall be responsible for the safe storage of material furnished by or to him, and accepted by him, and intended for the work until it has been incorporated in the completed project.

3.2 REMOVALS, TRENCH EXCAVATION, BACKFILL, AND RESTORATION

- A. Removal of existing facilities, trench excavation and backfill and restoration shall be in accordance with the provisions of these specifications.
- B. See Civil Improvement Plans.

3.3 WATER MAIN PIPE LAYING OPERATIONS

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies; and subject to compliance with the requirements of the Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.
- B. See Civil Improvement Plans.

3.4 TAPPING INTO THE WATERMAIN

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies; and subject to compliance with the requirements of the Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.
- B. See Civil Improvement Plans

3.5 METHODS OF TESTING AND DISINFECTING

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies; and subject to compliance with the requirements of the Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.

END OF SECTION

SECTION 33 12 00

WATER SERVICE CONNECTIONS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Water service connections; including meters, meter boxes, fire hydrants, and other appurtenances as indicated on the Drawings, specified herein, and required for a complete and proper installation.
 - 1. Service connections include connection to the distribution main, service line between main and the meter, meter with box, and service stops.

1.2 SUBMITTALS

- A. General:
 - 1. Submit under provisions of Section 01 33 00.
 - 2. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County, and respective Local Utility Companies.
- B. Product Data: Within 15 calendar days after the Contractor has received the Owner's Notice to Proceed, submit:
 - 1. Materials list of items proposed to be provided under this Section.
 - 2. Manufacturer's specifications and other data needed to prove compliance with the specified requirements.

1.3 QUALITY ASSURANCE

- A. Comply with the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County.
- B. Comply with the 2010 Uniform Design and Construction Standards for Potable Water Systems, 2nd Edition.
- C. Comply with the Uniform Standard Drawings for Public Work's Construction off-site Improvements, Nye County.
- D. Use adequate numbers of skilled workmen who are thoroughly trained and experienced in the necessary crafts and who are completely familiar with the specified requirements and the methods needed for proper performance of the work of this section.
- E. All materials in this section are to be manufactured in the United States.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, handle, and protect products to the site under provisions of Section 01 60 00.

1.5 FIELD MEASUREMENTS

- A. Verify actual locations of water distribution system and other construction to which water distribution must fit by accurate field measurements before installation. Coordinate installation schedule with construction progress to avoid delay of work.

1.6 WARRANTY

- A. Provide one (1) year written warranty under provisions of specifications.

Warranty: All materials and workmanship provided are guaranteed against defects after completion and final acceptance of the work. Defects due to faulty materials or workmanship developed during the guarantee period shall be satisfactorily repaired or replaced by the Contactor at his expense.

PART 2 - PRODUCTS

2.1 SERVICE PIPE

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- B. Subject to compliance with the requirements of the 2010 Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition; Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

2.2 BRASS MATERIALS

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- B. Subject to compliance with the requirements of the 2010 Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.
- C. Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

2.3 METER BOXES

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- B. Subject to compliance with the requirements of the 2010 Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.
- C. Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

2.4 METERS

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- B. Subject to compliance with the requirements of the 2010 Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.
- C. Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

2.5 OTHER MATERIALS

Provide other materials, not specifically described but required for a complete and proper installation, as selected by the Contractor subject to the approval of the Engineer.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine the areas and conditions under which Work of this Section will be performed. Correct conditions detrimental to proper and timely completion of the Work. Do not proceed until unsatisfactory conditions are corrected.

3.2 EXCAVATION AND BACKFILLING

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- B. Subject to compliance with the requirements of the 2010 Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.

- C. Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

3.3 INSTALLATION REQUIREMENTS

- A. Subject to compliance with the requirements of the Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- B. Subject to compliance with the requirements of the 2010 Uniform Design and Construction Standards for Potable Water Systems, 3rd Edition.
- C. Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

END OF SECTION

SECTION 33 30 00

SANITARY SEWER SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Sanitary sewer piping, fittings, accessories, cleanouts, and bedding
 - 2. Connection of site sanitary sewer system to municipal sanitary sewer system.

1.2 REFERENCES

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions, apply to this section
- B. Design and Construction Standards for Wastewater Collection Systems, 3rd Edition (DCSWCS).
- C. Uniform Standard Specifications for Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- D. Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County
- E. American Society for Testing and Materials (ASTM)
 - 1. ASTM D3034 – Type PSM Poly (Vinyl Chloride) (PVC) Sewer Pipe and Fittings

1.3 WORK SPECIFIED HEREIN

- A. Provide all labor, materials, equipment, and services necessary to complete the work consisting of furnishing and installing the complete sanitary sewer facilities, herein referred to as sewer laterals, excavation and backfill, preparation of pipe subgrade, construction of cleanouts, the furnishing, placing and testing of sanitary sewer facilities as shown on Construction Documents and in accordance with these specifications.
- B. Unless otherwise indicated on the plans and/or specified in these specifications, the construction of sewer laterals will include: excavation and backfill, cleanouts, supports, the furnishing, installing and testing of sanitary sewer pipelines, fittings, and all appurtenances; the removal and/or restoration of existing facilities, saw-cutting, the removal and/or replacement of AC paving, Type-II Base, etc. and all work in accordance with the plans and specifications.
- C. Subject to compliance with the requirements of the Uniform Standard Specifications for the Public Work's Construction off-site Improvements, Nye County and respective Local Utility Companies.
- D. Subject to compliance with the Standards and Specifications for Wastewater Collection Systems of Great Basin Water Company (Nye County).

- E. Subject to compliance with the Uniform Standard Drawing for Public Work's Construction off-site Improvements, Nye County.

1.4 SUBMITTALS

- A. Contractor shall submit all specified product literature and shop drawings in accordance with the requirements of these specifications.
- B. Provide County's representative with data on pipe materials, pipe fittings, and accessories.
- C. Manufacturer's Certificate: Certify that products meet and/or exceed state and local requirements.
- D. As-Builts: Accurately record actual installed locations of piping mains, cleanouts, connections, top of pipe elevations and clearances to existing utilities.
- E. Identify and describe unexpected variations to subsoil conditions and location of uncharted utilities.

1.5 EXISTING UNDERGROUND UTILITIES

- A. Existing underground sewer lines, water mains, drainage facilities, and dry utilities shown on the Construction Documents have been located per field survey, as-built and design documents and field investigations. The County presumes no responsibility as to their location.
- B. Contractor will verify the invert elevation and pipe size of the existing sanitary sewer line prior to installation of any on-site facilities and notify the county representative and Civil Engineer of any discrepancies to the construction documents.
- C. All existing public and private sanitary sewer lines will remain operational during construction activities. Temporary diversions or service interruptions are not allowed.
- D. Contractor will verify utility and storm drain crossing elevations as noted on the Construction Documents and notify county representative and Civil Engineer of any discrepancies.

1.6 QUALITY ASSURANCE

- A. All products, construction materials and methods to be used or incorporated into the work shall be subject to the inspection and approval or rejection of the county representative. The materials to be used will be new and meet all quality standards as defined by these standards.

PART 2 - PRODUCTS

2.1 PRODUCTS

- A. Polyvinyl Chloride Pipe (PVC): ASTM D3034, rated SDR 35 unless otherwise specified by Clark County. Pipes shall be continually marked with manufacturer's name, pipe size, cell classification, SDR rating, and ASTM D3034 classification.
- B. Fittings will be same material as pipe molded or formed to suit pipe size and end design in required bends, wyes, cleanouts, traps, etc.
- C. Magnetic detectable conductor (#12 copper) trace wire, brightly covered plastic covering, imprinted with "Sanitary Sewer Service" in large letters.
- D. See Civil Construction Documents.

PART 3 - EXECUTION

3.1 EXECUTION

- A. All work shall be in accordance with DCSWCS, 3rd Edition.
- B. Verify that service connections and municipal utility sanitary sewer main size, invert elevations, location, and depth are as indicated on Construction Documents.
- C. Install trace wire continuous over top of pipe.
- D. Maintain separation of potable water main from sewer piping at crossings a minimum of 18-inches vertical and 10-feet horizontal.
- E. Hand trim and form bottom of excavations clean and smooth to required elevations. Correct over excavations with bedding material.

END OF SECTION

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